Essays

|!|BCEEC0130000001|!|
Describe how flowering plants can reproduce sexually by means of flowers and seeds.
(12 marks)

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Content	Description	Mark
Pollination	- Male gametes are produced in the anthers by meiotic cell division. - Male gametes are carried by pollen grains. - Pollen grains are transferred from the anther to the stigma by means of insects / wind / external agents.	(Max. 2)
Fertilization	- When a pollen grain lands on the stigma, the sugary solution secreted by the stigma stimulates the pollen grain to develop a pollen tube. - The pollen tube secretes enzymes to digest the tissues of the carpel, and grows down through the style and the ovary. - The pollen tube then enters the micropyle and releases the male gamete. - The male gamete then fuses with the female gamete to form a zygote.	(Max. 3)
Development of fruits and seeds	- After fertilization, the ovary gradually develops into a fruit. - The zygote develops into the embryo. - The ovule develops into the seed. - Integuments develop into the seed coat. - The ovary wall develops into the fruit wall.	(Max. 2)
Development of a new plant	- The fruits / seeds are dispersed by wind / animal / water / other reasonable answer. - Under favourable conditions (e.g. optimum temperature, with adequate water), the seeds germinate. - The embryo in the seed uses the food reserves in cotyledons to develop into a new plant.	(Max. 2)
Communication		(Max. 3)

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(12 marks)

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Skill assessed	Marks
Understanding basic concepts	9
Communication	3

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